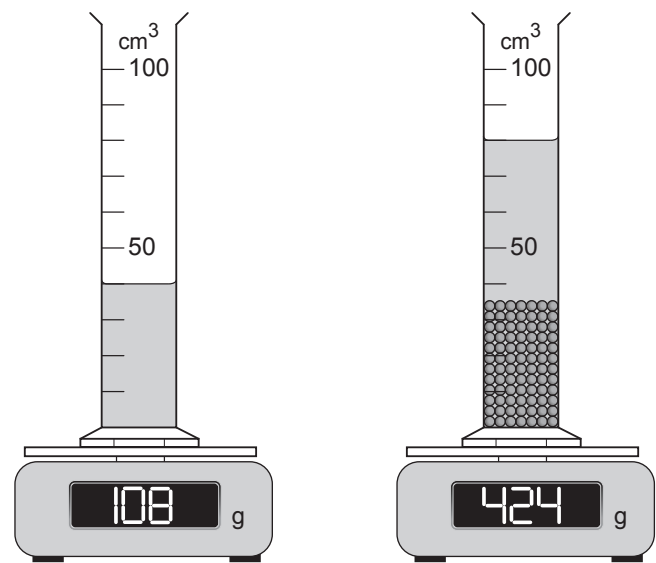


4. A group of students investigate density using an electronic balance, a measuring cylinder, water and some small steel ball bearings.
- They set the balance to zero and place a measuring cylinder on to the balance.**
- They then add water at 10 cm³ intervals and record both the mass and volume.
- At some point they start adding the steel ball bearings instead of water, and record the mass and volume each time the water level rises by 10 cm³.
- They continue until the final volume of 100 cm³ is reached.
- The arrangement is shown below.



The students produce a table of their results, which is shown below.

Volume (cm ³)	Mass (g)
0	68
10	78
20	88
30	98
40	108
50	187
60	266
70	345
80	424
90	503
100	582



Examiner
only

- (a) One member of the group claims that water has a density value of 1 g/cm^3 . Explain whether her claim is correct. [No calculations are required.] [1]

.....

.....

- (b) (i) State the volume at which they start adding the steel ball bearings. [1]

Volume = cm^3

- (ii) Give a reason for your answer. [1]

.....

.....

- (c) Use an equation from page 2 to calculate the density of steel. [2]

Density of steel = g/cm^3

- (d) State **two** improvements that could be made to this experiment that would lead to a more accurate value for the density of steel. [2]

1.
2.

7

